

Design FMEA (qualitative) - GridStore-D3

VBF-T2R1-DFMEA-01 | t2_r1 grid energy storage | 2026-08-25 | values mechanically sourced from parameter set / simulation outputs

Failure-mode table (qualitative; S/O high/medium/low based on simulation signals)

#	Failure mode	Failure cause	Simulation signal	S	O	Mitigation recommendation
1	Anode lithium plating at 4C	transport-limited anode polarization at 4C/45C	anode_potential_v min +0.0498 V (thinnest margin in family; baseline -0.1918 V)	high	low	keep transport formulation (sigma 3.0, D 2.5e-9, r_n 2.0um, r_p 2.5um); margin monitoring; fallback D1 (+0.091 V) on record
2	Excessive temperature rise at fast charge	self-heating at 4C	T_max 342.39 K at 318.15 K ambient (+24.2 K)	medium	medium	no contract red line; verify cell-level cooling in field use (cooling h not a design lever here)
3	Electrolyte oxidative decomposition at high voltage	cathode potential vs electrolyte stability	window capped 4.2 V; HOMO endorsement skipped (real_compute=false)	medium	low	keep 4.2 V cutoff; run Stage 2 molecular validation before production (honest gap)
4	Insufficient capacity / ED	loading / transport mismatch	1C 5.066 Ah >= 5.0; ED 465.62 >= 327.18	low	low	none required (margin x1.42)
5	Low-temperature capacity loss	electrolyte transport freezing at -20C	retention 99.571% (protocol + true-soak, 266.7 K)	low	low	constant-transport formulation proxy is the enabler; validate real electrolyte at -20C physically
6	Long-term SEI overgrowth (>550 nm @500cyc)	SEI growth kinetics	25.32 nm @500 cyc (k_sei 1e-16 coating bridge; baseline 777.9 nm)	medium	low	actual coating must be developed and validated (Stage 2 + physical aging); without it baseline kinetics FAIL
7	Model bookkeeping inconsistency (porosity override)	active vf (0.665/0.75) defined at base porosity; 0.40 makes active+pore>1	parameter-set arithmetic (design_spec section 6)	low	low	physical electrode reformulation (binder/additive accounting) before manufacture; annotated in spec

Conclusion: highest-risk items are #1 (plating margin thin at +0.0498 V) and #6 (SEI-kinetics rests on the coating bridge - physical validation mandatory). Mitigations embedded in design; fallback designs on record. Process/supplier-level FMEA: N/A (beyond pure simulation boundary).